

Standard Operating Procedure & Troubleshooting Manual

Equipment: Injection Moulding Machine IM-105 (250 t)

Document Ref: SOP-ENG-IM105-REV4

Plant Location: Moulding Hall, Bay 3

Applicability: Process Operators, Maintenance Technicians, Shift Supervisors

1. Equipment Description

Machine **IM-105** is a 250-tonne hydraulic injection moulding machine producing a polypropylene (PP) container lid in a single-cavity mould with a hot runner. Specified part weight is 48.0–49.0 g. A screw with a non-return (check) ring injects the melt; the **cushion** is the melt left in front of the screw at the end of injection and shows whether the shot size is adequate. Mould temperature is controlled by a chiller through four cooling circuits.

2. Operating Envelope & Alarm Limits

Table 1: Operating envelope and alarm limits

Parameter	Normal Range	Warning	Critical Alarm	Trip / Action
Melt temperature	220 – 240 °C	above 245 °C or below 215 °C	above 250 °C or below 210 °C	260 °C: barrel heater cut-off
Peak injection pressure	900 – 1,200 bar	above 1,300 bar	above 1,400 bar	1,500 bar: machine limit
Mould temperature	30 – 45 °C	above 50 °C	above 55 °C	Check chiller
Cushion	4 – 8 mm	below 3 mm or above 10 mm	below 1 mm	Stop and check shot size / check ring
Cycle time	18 – 22 s	above 24 s	above 28 s	Supervisor review
Part weight	48.0 – 49.0 g	below 47.8 g or above 49.2 g	below 47.5 g or above 49.5 g	Quarantine parts

Part weight is checked by the operator every 30 minutes and recorded on the shift sheet.

3. Fault Matrix

Table 2: Faults, signs, causes and actions

Fault	Typical Signs	Likely Causes	Operator Action
FAULT-A: Short shot	- Part incomplete, weight low - Cushion near zero, or injection pressure at limit	Shot size too small, worn check ring (cushion drifts to zero), melt too cold, blocked gate or vents.	1. Check cushion. 2. Increase shot size in small steps. 3. Check melt temperature. 4. Inspect gate and vents. 5. Hold parts made since the last good check.
FAULT-B: Flash	- Thin film at the parting line - Part weight high	Clamp force low, injection or hold pressure too high, melt too hot, damaged parting line.	1. Check actual clamp force. 2. Reduce hold pressure in steps. 3. Check melt temperature. 4. Request mould inspection.
FAULT-C: Sink marks / voids	- Depressions on thick sections - Part weight slightly low - Cushion NORMAL	Hold pressure or hold time too low, mould or melt too hot.	1. Increase hold pressure or hold time. 2. Check mould temperature and chiller.
FAULT-D: Warpage	Lid not flat, dimensions out of tolerance	Uneven mould cooling (blocked circuit), cooling	1. Check water flow in each cooling circuit.

	• Weight NORMAL	time too short, mould too hot.	2. Increase cooling time. 3. Compare mould temperature of both halves.
--	-----------------	--------------------------------	---

4. Operating Rules

- Change one setting at a time and record every change on the shift sheet.
- Quarantine all parts made since the last good weight check when a fault is found.
- Never reach into the mould area unless the safety gate is open and the machine is locked out.
- Settings outside the operating envelope need process engineer approval.